

Peroxide-Catalyzed Swell Grafting of Maleic Anhydride Onto Polypropylene

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A new grafting method was developed to incorporate maleic anhydride directly onto solid-state polypropylene powders. Maleic anhydride grafts altered the nonpolar characteristics of polypropylene so that much better mixing was achieved in blends and composites of polypropylene with many other polymers and fillers. Maleic anhydride was grafted onto polypropylene by the peroxide-catalyzed swell grafting method, with a maximum extent of grafting of 4.60%. Fourier transform infrared spectroscopy, differential scanning calorimetry, thermogravimetric analysis, scanning electron microscopy, tensile testing, and impact testing were used to characterize the isotactic polypropylene (iPP), maleic anhydride grafted polypropylene (MAH-*g*-iPP), and (isotactic polypropylene)/(calcium carbonate) composites (iPP/CaCO₃). The crystallinity and heat of fusion of the MAH-*g*-iPP decreased as the extent of grafting increased. The mechanical properties of the CaCO₃ filled polypropylene were improved by adding MAH-*g*-iPP as a compatibilizing agent. The dispersion of the fillers in the polymer matrix and the adhesion between the CaCO₃ particles and the polymer matrix were improved by adding the compatibilizer.

INTRODUCTION

Isotactic polypropylene (iPP) is one of the fastest growing major thermoplastics. The current worldwide production of this resin is well over 20 billion pounds, and the growth rate is greater than those of many other commodity polymers (1). Polypropylene is readily available at a relatively low cost and yet possesses many unique properties that are suitable for a wide variety of consumer and industrial products. It is light in weight, flexible, and has excellent chemical resistance and a high melt temperature. These properties contribute to the broad range of applications of polypropylene in injection molding, films, and filaments.

Polypropylene is a semicrystalline thermoplastic. Its nonpolar property makes it very difficult to combine with many other polymers and fillers. The adhesion properties with the other materials and the printing properties are also influenced by its nonpolar characteristics. In order to improve its adhesion properties,

it is an efficient method to graft some unsaturated monomers such as maleic anhydride (MAH) or acrylic acid to the polypropylene chains (2–5). Usually, these unsaturated monomers are grafted to the polypropylene chains by either solution or melt processing (6–13). Both of these grafting methods possess disadvantages. For example, it is very difficult to completely remove all the solution and monomer residues from the grafted polypropylene.

In this investigation, a new swell grafting method to graft maleic anhydride onto solid-state polypropylene has been developed. The grafting reaction is carried out on the surface of polypropylene particles by adding some surface activating agents and peroxides. When the grafting reaction is completed, it is rather easy to remove the monomer residues and the small amount of solution from the solid-state grafted polypropylene. Therefore, this peroxide-catalyzed swell grafting method is much more economical than the traditional solution grafting method and the extent of grafting is higher than that obtained from the melt grafting method. This maleic anhydride grafted polypropylene (MAH-*g*-iPP) was added to the inorganic

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filler [i.e. calcium carbonate (CaCO_3)] filled polypropylene as a compatibilizer. The tensile and impact strength of the composite were improved greatly.

This paper discusses the thermodynamic behavior of MAH-*g*-iPP and the effects of MAH-*g*-iPP as a compatibilizer on the mechanical properties of CaCO_3 filled polypropylene (iPP/ CaCO_3 composites). The grafting reaction mechanism, which has been reported in many other publications (14–16), will not be described.

EXPERIMENTAL

Preparation of Maleic Anhydride Grafted Polypropylene

The reaction of maleic anhydride with isotactic polypropylene (iPP) was carried out in a four-neck round-bottom flask. 100 g of polypropylene powder (Himont U.S.A., product no.: 6201, MFI = 20), 10 g of surface activating agent [xylene (Aldrich, reagent grade) or toluene (Aldrich, reagent grade)], and 3, 5, 10, 15 or 30 g of maleic anhydride (Aldrich, 99%) were mixed together in the reaction flask to prepare samples of various graft levels for thermal analyses. The speed of agitation was about 90 rpm. The temperature was maintained at 60°C for 90 min in order to make the polypropylene particles swell up. A xylene solution of peroxide [1 g of benzoyl peroxide (Aldrich, 97%) dissolved in 10 g of xylene] was dropped into the reaction flask in 20 min. The reaction was maintained at 100°C for 3 h. After the reaction was completed, the product (MAH-*g*-iPP) was washed with a large amount of warm water until pH = 7. Finally, the MAH-*g*-iPP was washed with acetone (Aldrich, reagent grade) and dried at 70–80°C for 24 h in vacuum. The MAH (2%)-*g*-iPP compatibilizer was prepared with 8 g of maleic anhydride. All chemicals were used as received without further purification.

Extent of Grafting

The MAH-*g*-iPP sample was extracted for 3 h with acetone in a Soxhlet extraction apparatus, filtrated and dried for 24 h in vacuum. The extent of grafting was calculated by the following equation:

% Grafting (G) =

$$\frac{\text{Weight of MAH-}g\text{-iPP} - \text{Weight of iPP}}{\text{Weight of iPP}} \times 100\%$$

A control iPP sample was prepared with the identical experimental procedures for MAH-*g*-iPP samples except no maleic anhydride was added. It was found that the control iPP sample did not contain any acetone extractable components since there was no detectable weight change after the swelling, washing, and drying process.

Preparation of Polypropylene/ CaCO_3 Composites

Eight iPP/ CaCO_3 composites with different weight ratios of the MAH-*g*-iPP compatibilizer were prepared.

The weight ratio of iPP/ CaCO_3 was a constant of 4:1 for all the samples. The compatibilizer used was MAH-*g*-iPP with 2.00% extent of grafting [MAH (2%)-*g*-iPP] and it was added to the iPP/ CaCO_3 mixtures in the following weight ratios: 0 phr, 1.0 phr, 1.5 phr, 2.0 phr, 2.5 phr, 3.0 phr, 3.5 phr, and 4.0 phr. The composites were prepared by Haake Rheocord 9000 Counter-Rotating Twin Screw Extruder operated at 20 rpm, and the thermal profile of the feed, transition, metering and die sections of the extruder was set at 170°C, 180°C, 180°C, and 180°C, respectively. The components were pre-mixed mechanically in a laboratory blender before they were fed into the hopper of the extruder. A thermal stabilizer (Ciba-Geigy Irganox 1010) was also added to the extruder in the weight ratio of 0.5 part per hundred parts of iPP resin. The components were then blended in the extruder and formed into pellets.

Thermal Analyses

Differential scanning calorimetry (DSC) was performed with a TA Instruments DSC 2910 Differential Scanning Calorimeter under a nitrogen atmosphere in order to investigate the effects of the extent of grafting of MAH-*g*-iPP on its crystallization and melting behavior. Initially, the temperature of the samples was raised from 30°C to 200°C at a rate of 10°C/min in order to impart a uniform thermal history to all of the samples. Then the samples were cooled from 200°C to 30°C at a rate of -10°C/min and the crystallization curves were taken. Afterward, the samples were heated again from 30°C to 200°C at a rate of 10°C/min and the melting curves were taken at this time.

The thermal stability of iPP and MAH-*g*-iPP was studied by thermogravimetric analysis (TGA) in air with TA Instruments TGA 2950 Thermogravimetric Analyzer. The temperature of the MAH-*g*-iPP samples was raised from 30°C to 550°C at a rate of 10°C/min and their thermograms were recorded.

Fourier Transform Infrared Spectroscopy

Fourier transform infrared spectroscopy (FTIR) was used to confirm the grafting of MAH onto iPP in the MAH-*g*-iPP. The samples were pressed KBr pellets in which iPP or MAH-*g*-iPP powders were dispersed. The FTIR spectra were obtained by Perkin Elmer 16PC FTIR Spectrometer.

Mechanical Properties

Tensile modulus, tensile strength at break, and unnotched Izod impact strength of the iPP/ CaCO_3 composites were obtained in order to find out the effects of compatibilizer on the mechanical properties of the composites. First, the composite materials were pressed into 4-mm-thick sheets by PHI Hydraulic Hot Press operated at 180°C and 25 ton for 10 min. Then, test specimens were cut from the sheets according to the standard sizes specified by the ASTM test methods. The tensile tests were performed on MTS Sintech

10/D Universal Tensile Tester according to ASTM Test 638 at 20°C and crosshead speed of 50 mm/min. The Izod impact tests of unnotched samples were performed on Ceast 6545 Universal Pendulum Impact Tester according to ASTM Test D256.

Scanning Electron Microscopy

Scanning electron microscopy (SEM) was used to investigate the influence of MAH-*g*-iPP as a compatibilizing agent on the compatibility of CaCO₃ with polypropylene and the dispersion of CaCO₃ particles in the iPP/CaCO₃ composites. The experiments were carried on JEOL JSM-6300 Scanning Electron Microscope. The samples used were the same specimens from the tensile test experiments.

RESULTS AND DISCUSSION

Differential Scanning Calorimetry

The DSC cooling thermograms of iPP and MAH-*g*-iPP samples are shown in Fig. 1, performed at a cooling rate of 10°C/min. The crystallization temperature (T_c) for iPP was 118°C and the heat of crystallization (ΔH_c) was 101 J/g (Fig. 2). A shoulder is observed in the crystallization peak. It is probably due to the formation of imperfect crystallites at a lower temperature when the iPP was subjected to the relatively fast cooling process in this experiment. The crystallization temperatures of MAH-*g*-iPP were 112°C, 114°C, 113°C, 113°C, and 114°C with increasing extent of grafting and their heat of crystallization were 99 J/g,

98 J/g, 94 J/g, 89 J/g, and 84 J/g, respectively (Fig. 2). The temperature of crystallization and the heats of crystallization decreased after the grafting reactions. In addition, the heat of crystallization, which indicates percent crystallinity of MAH-*g*-iPP, decreased as the extent of grafting increased. The shoulder on the crystallization peak of iPP also disappeared in the MAH-*g*-iPP. It was probably due to the grafting reaction affecting the imperfect crystallites of iPP just as well as the amorphous phase of iPP. The decrease in crystallinity after chemical grafting was probably caused by the increase in difficulty of polymer chain arrangements as the grafted branch chains prohibited the movements of the polymer segments (17). The results indicated that the grafting reaction occurred not only on the surfaces of the polymeric particles, but also in the cores of these particles.

DSC heating thermograms of iPP and MAH-*g*-iPP samples are shown in Fig. 3. The melting temperature (T_m) of iPP (164°C) was slightly higher than those of MAH-*g*-iPP (162–163°C). And, the heat of fusion (ΔH_f) of iPP (107 J/g) was also slightly higher than those of MAH-*g*-iPP, which were respectively 105 J/g, 103 J/g, 100 J/g, 96 J/g, and 90 J/g in the order of increasing extent of grafting (Fig. 2). The results showed that the heat of fusion decreased as the extent of grafting increased. It was probably due to the grafted branches, which disrupted the regularity of the chain structures in iPP and increased the spacings between the chains; consequently, the percent crystallization, and therefore heat of fusion, decreased.

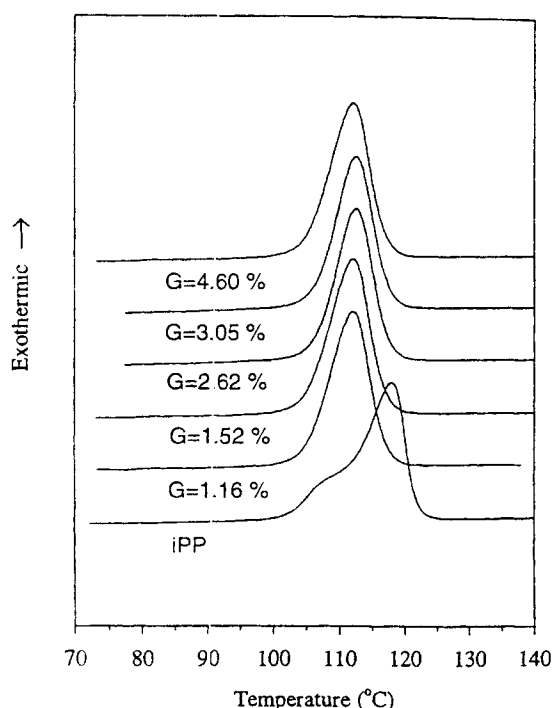


Fig. 1. DSC cooling thermograms of iPP and MAH-*g*-iPP.

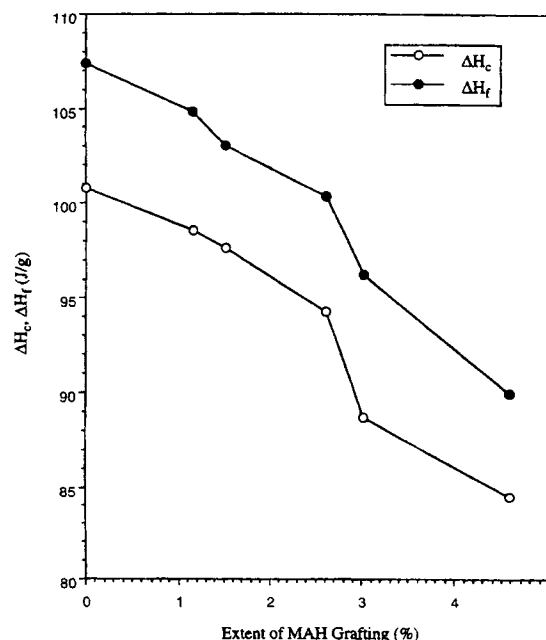


Fig. 2. Heat of crystallization (ΔH_c) and heat of fusion (ΔH_f) of iPP and MAH-*g*-iPP.

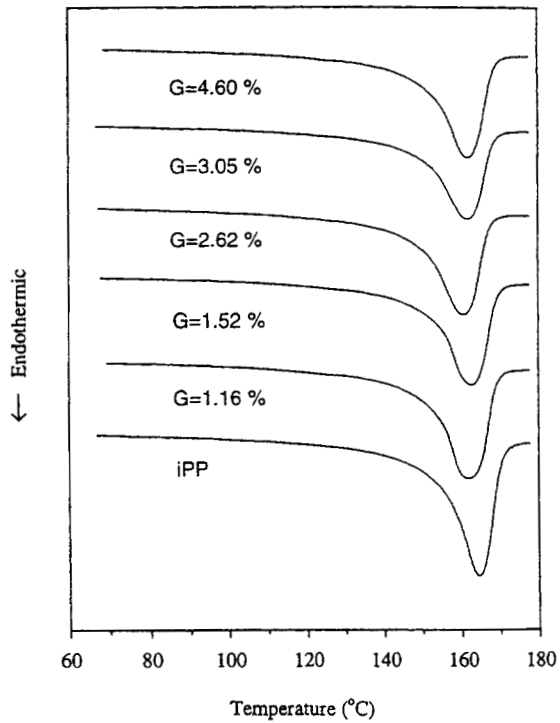


Fig. 3. DSC heating thermograms of iPP and MAH-g-iPP.

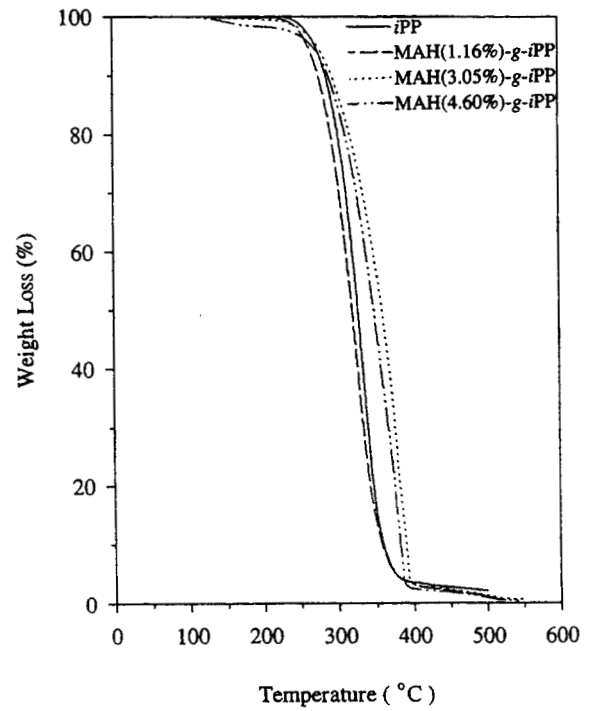


Fig. 4. TGA thermograms of iPP and MAH-g-iPP.

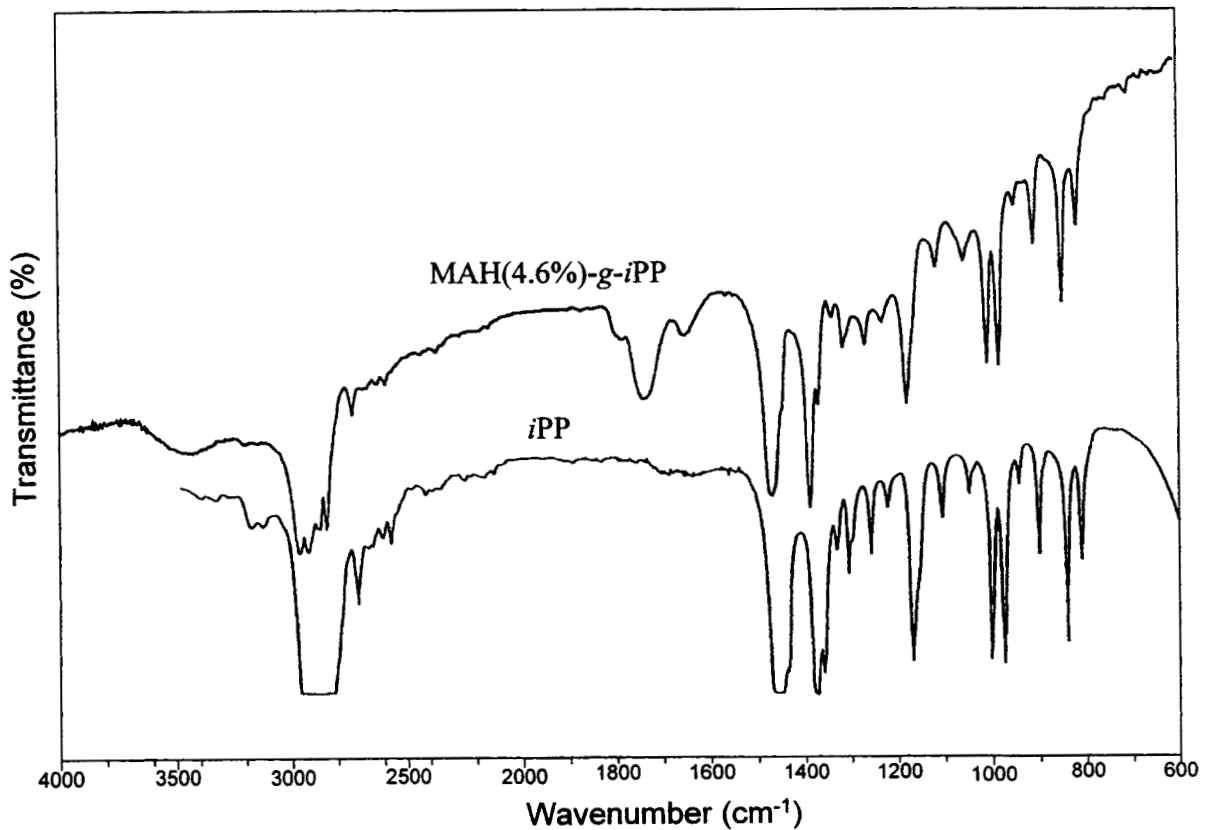


Fig. 5. FTIR spectra of iPP and MAH(4.6%)-g-iPP.

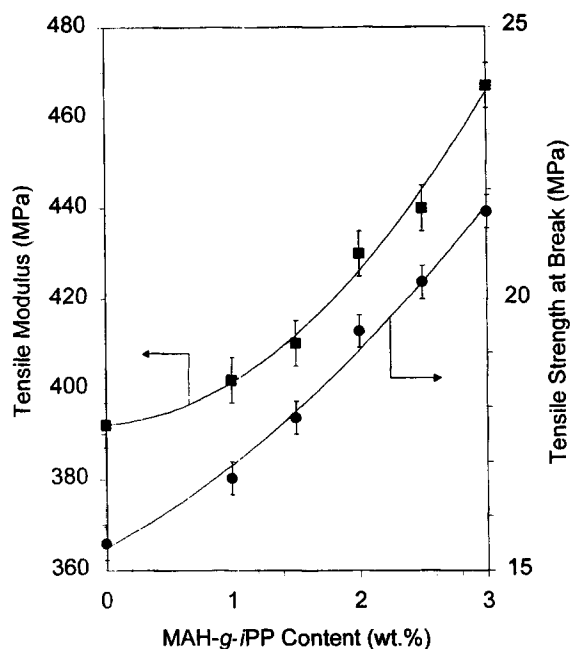


Fig. 6. Effects of compatibilizer [MAH (2%)$-g$-iPP] on the (a) tensile modulus and (b) tensile strength at break of iPP/CaCO₃ composite.

Thermogravimetric Analysis

Figure 4 shows the TGA results. All samples showed an onset temperature of decomposition at about 210–220°C and were almost completely decomposed at above 400°C. There were no significant changes in thermal stability after the grafting reactions.

Fourier Transform Infrared Spectroscopy

The FTIR spectra of iPP and MAH-*g*-iPP are shown in Fig. 5. In the MAH-*g*-iPP FTIR spectrum, a strong characteristic peak is revealed at 1732 cm⁻¹ that indicated the presence of the carboxylic groups in the MAH-*g*-iPP. In contrast, in the iPP FTIR spectrum, there is no indication of the presence of any peaks in the range between 1600 and 2000 cm⁻¹. This result clearly showed that the maleic anhydride monomers had been successfully grafted onto the polypropylene chains in the swell grafting process.

Mechanical Properties

Figures 6 and 7 show the tensile moduli, tensile strength at break, and unnotched Izod impact strength of the iPP/CaCO₃ composites. All of these mechanical properties were greatly improved when the MAH (2%)$-g$-iPP compatibilizer was added to the iPP/CaCO₃ composites. The compression molded iPP/CaCO₃ composite sample containing no compatibilizers showed a tensile modulus value of 392 MPa, which was much lower than those of typical unfilled iPP homopolymer injection molded T-bars of 1.1–1.5 GPa (18). The tensile modulus and the tensile strength at break increased monotonically from 392 MPa to 467 MPa and

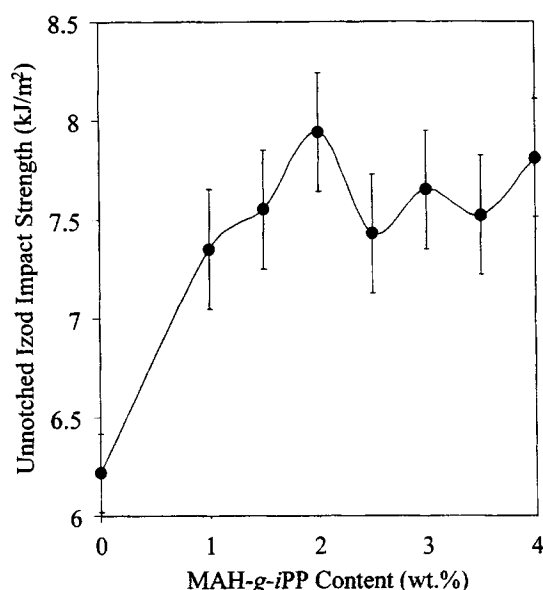
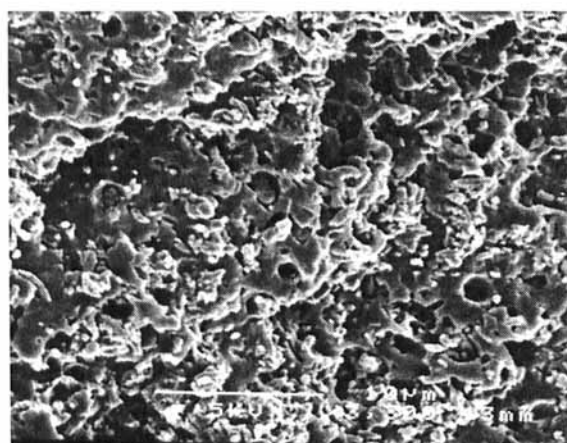


Fig. 7. Effect of compatibilizer [MAH (2%)$-g$-iPP] on the unnotched Izod impact strength of iPP/CaCO₃ composite.

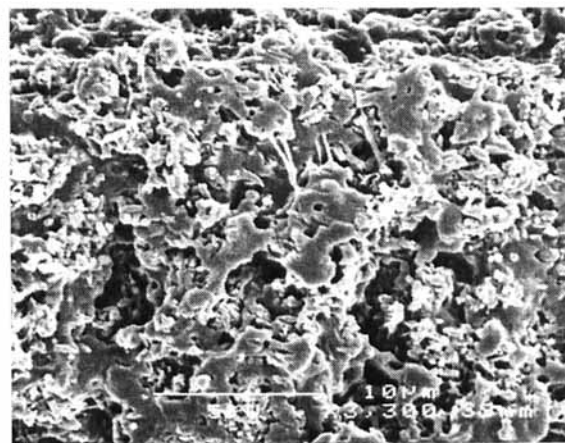
from 15.5 MPa to 21.6 MPa, respectively, when the compatibilizer content increased from 0 wt% to 3.0 wt% (Fig. 6). The Izod impact strength of unnotched samples was improved from 6.2 ± 0.2 kJ/m² to the range of 7.4 ± 0.3 kJ/m² to 7.9 ± 0.3 kJ/m² when the compatibilizer was added. The data scattered within the error range of the measurements. The average Izod impact strength of the unnotched iPP/CaCO₃ composite samples was 7.6 ± 0.3 kJ/m² after MAH (2%)$-g$-iPP was added. The bulk mechanical properties of (semicrystalline polypropylene)/(inorganic filler) composites usually depend to a large extent on the dispersion of the filler particles in the composites and the interactive forces between the polymer matrices and the filler particles. Although the dispersion is related to the thermodynamic compatibility of the components and the processing conditions, this thermodynamic compatibility is usually controlled by a third component, i.e. a compatibilizer. The compatibilizer is able to mediate an attractive interaction between the phases (18–20). In this study, all of the mechanical properties were improved by the addition of a compatibilizer with polar functional groups. This indicated that the introduction of polar functional groups on the polymer chains improved adhesion between the polymer matrix and the filler particles, and hence resulted in more efficient stress transfer at the polymer/filler interface (21).

Scanning Electron Microscopy

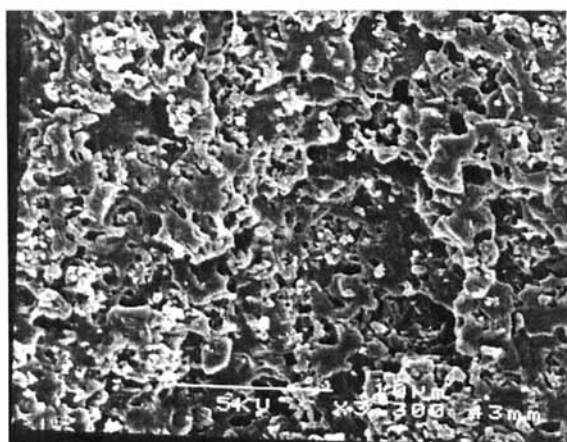
Figure 8 shows the SEM micrographs of iPP and iPP/CaCO₃ samples. For the iPP/CaCO₃ composite without compatibilizer, Fig. 8a shows that the CaCO₃



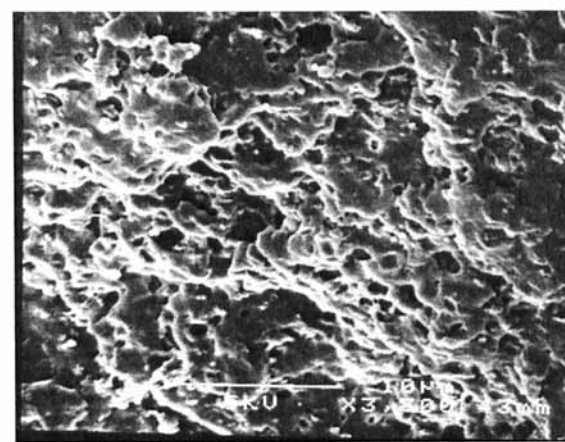
(a) 0 wt%



(b) 1 wt%



(c) 2 wt%



(d) 3 wt%

Fig. 8. SEM micrographs of the iPP/CaCO₃ composite with different weight ratios of the compatibilizer [MAH (2%)-*g*-iPP]: (a) 0 wt%, (b) 1.0 wt%, (c) 2.0 wt%, and (d) 3.0 wt%.

fillers existed in the form of large aggregates of particles. The interfaces between the polymer matrix and the CaCO₃ particles were clearly defined. The difference of surface tension between the polymer matrix and the filler particles was so large that it was very difficult to combine the polymer and filler together. In contrast, Figs. 8b-d show that the dispersion of the CaCO₃ particles was much improved in the polypropylene matrix when the MAH (2%)-*g*-iPP compatibilizer was added to the iPP/CaCO₃ composite. In addition, the SEM micrographs also indicated that as the weight ratio of the MAH (2%)-*g*-iPP compatibilizer in the composite increased, both of the dispersion of the CaCO₃ fillers and the polymer/filler interfacial interactions were enhanced. The boundaries between the polymer matrix and the filler particles became unclearly defined and eventually the CaCO₃ particles were completely wrapped up by the polymer matrix as the relative amount of the compatibilizer increased. Simultaneously, the mechanical properties were also greatly improved as the compatibilizer content

increased, which was discussed above. These results indicated that it is very difficult to obtain a uniform dispersion and the desired material properties when mixing polymer and filler particles together without using a suitable compatibilizing agent. In this study, MAH (2%)-*g*-iPP was used as the compatibilizer in the composite of CaCO₃ filled iPP in order to enhance the adhesion between iPP and CaCO₃ particles, and hence a finer dispersion and improved mechanical properties.

CONCLUSIONS

The experimental results of this work indicated that the swell grafting of maleic anhydride onto polypropylene powders catalyzed by benzoyl peroxide was successful. FTIR spectra showed a strong characteristic peak for the MAH-*g*-iPP. DSC analyses showed that the crystallinity and the heat of fusion of MAH-*g*-iPP decreased as the extent of grafting increased. TGA results indicated that the extent of grafting has no significant effects on the thermal stability properties of

iPP. The mechanical properties of CaCO₃ filled iPP were improved by using MAH-*g*-iPP as the compatibilizing agent. The impact strength was increased by about 23% after the MAH (2%)-*g*-iPP compatibilizer was added. The dispersion and adhesion of the CaCO₃ filler particles in the iPP polymer matrix were also improved after adding the MAH-*g*-iPP compatibilizing agent as shown in the SEM micrographs.

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